

DWV Assignment 7 Swiston

Bryn Swiston

```
##Bryn Swiston Assignment 7 #Question 1
```

```
#Set Working Directory  
setwd("~/Documents/Data Wrangling Library/Week 7 Homework Files")
```

```
#Open and Activate the Appropriate Libraries  
library(haven)  
library(tidyverse)
```

```
Warning: package 'ggplot2' was built under R version 4.5.2
```

```
Warning: package 'tibble' was built under R version 4.5.2
```

```
Warning: package 'tidyr' was built under R version 4.5.2
```

```
Warning: package 'readr' was built under R version 4.5.2
```

```
Warning: package 'purrr' was built under R version 4.5.2
```

```
Warning: package 'dplyr' was built under R version 4.5.2
```

```
Warning: package 'lubridate' was built under R version 4.5.2
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --  
v dplyr      1.2.1      v readr      2.2.0  
v forcats    1.0.1      v stringr    1.6.0  
v ggplot2    4.0.3      v tibble     3.3.1  
v lubridate  1.9.5      v tidyr      1.3.2  
v purrr      1.2.2
```

```
-- Conflicts ----- tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag() masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
#Read in the Cumulative American National Election Studies (ANES) File
mydata <- read_dta('/Users/brynswiston/Documents/Data Wrangling Library/Week 7 Homework Files')
```

```
#Filter Data by Variable of Interest and Year, VCF0301 and VCF0004
PartyID <- c("VCF0004","VCF0301")
```

```
#Create Data Set for the 2 Variables of Interest
PartyIDyearly <- mydata[, PartyID]
```

```
#Filter Data by Years Used in Fig 1
PartyID52to96 <- PartyIDyearly |> filter(VCF0004%in%1952:1996)
```

```
#Remove the misspelled PartIDYearly dataframe because it's bothering me
rm(PartIDyearly)
```

Warning in rm(PartIDyearly): object 'PartIDyearly' not found

```
#Recode the PartyID Identifiers so We Can Get Either Strong or Weak, Pure or Lean
PartyID52to96_r <- PartyID52to96|>
  mutate(VCF0301_r = case_when(
    VCF0301 %in% c(1, 7) ~ "strong",
    VCF0301 %in% c(2, 6) ~ "weak",
    VCF0301 %in% c(3, 5) ~ "lean",
    VCF0301 %in% c(4) ~ "pure"
  ))
```

```
#Double Check the Counts with a Table
table(PartyID52to96_r$VCF0301_r, useNA = "ifany")
```

lean	pure	strong	weak	<NA>
8244	4768	12262	14851	840

```
table(PartyID52to96$VCF0301, useNA = "ifany")
```

```

  0    1    2    3    4    5    6    7
840 7704 9059 4393 4768 3851 5792 4558

```

```
#Filter Out the NA Data
```

```
NoNAPartyID52to96_r <- PartyID52to96_r |>
  filter(!is.na(VCF0301_r))
```

```
#The change in the number of observations matches the NAs identified in the earlier table, so
```

```
#Group by Year and Category, then Summarise and Find Proportions
```

```
yearlyprop <- NoNAPartyID52to96_r |>
  group_by(VCF0004, VCF0301_r) |>
  summarise(countID = n()) |>
  mutate(prop = countID / sum(countID))
```

```
`summarise()` has regrouped the output.
```

```
i Summaries were computed grouped by VCF0004 and VCF0301_r.
```

```
i Output is grouped by VCF0004.
```

```
i Use `summarise(.groups = "drop_last")` to silence this message.
```

```
i Use `summarise(.by = c(VCF0004, VCF0301_r))` for per-operation grouping
  (`?dplyr::dplyr_by`) instead.
```

```
#Create New Columns Grouping by Party or Independent for the Two Graphs
```

```
yearlyprop <- yearlyprop |>
  mutate(dualgraph = case_when(
    VCF0301_r %in% c("strong", "weak") ~ "party",
    VCF0301_r %in% c("lean", "pure") ~ "independent"
  ))
```

```
#Try to Plot the Two Graphs
```

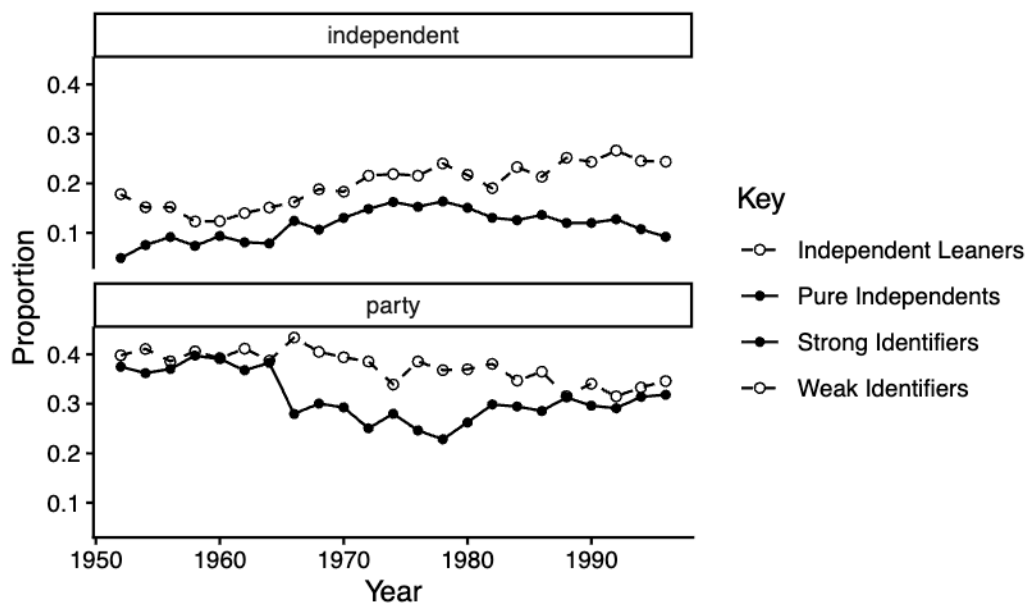
```
ggplot(yearlyprop, aes(x = VCF0004, y = prop, group = VCF0301_r, linetype = VCF0301_r, shape =
  labs(x = "Year",
    y = "Proportion",
    title = "Figure 1: The Distribution of Party Identification, 1952-1996") +
  facet_wrap(~ dualgraph, ncol = 1) +
  scale_shape_manual(
    values = c("strong" = 16, "pure" = 16, "weak" = 1, "lean" = 1),
    labels = c("lean" = "Independent Leaners", "pure" = "Pure Independents", "strong" = "Str
```

```

    name = "Key"
  ) +
  scale_linetype_manual(
    values = c("strong" = "solid", "pure" = "solid", "weak" = "dashed", "lean" = "dashed"),
    labels = c("lean" = "Independent Leaners", "pure" = "Pure Independents", "strong" = "Str
    name = "Key"
  ) +
  theme_classic() +
  geom_line() +
  geom_point()

```

Figure 1: The Distribution of Party Identification, 1952–1996



```

#I Think I have to Build Two Seperate Graphs...
indep_data <- yearlyprop |> filter(dualgraph == "independent")
party_data <- yearlyprop |> filter(dualgraph == "party")

#Build the graph (with the independent data) that will go on the bottom
bottom_ggplot <- ggplot(indep_data, aes(x = VCF0004, y = prop, group = VCF0301_r, linetype =
  labs(x = "Year",
    y = "Proportion") +
  scale_shape_manual(
    values = c("strong" = 16, "pure" = 16, "weak" = 1, "lean" = 1),
    labels = c("lean" = "Independent Leaners", "pure" = "Pure Independents", "strong" = "Str

```

```

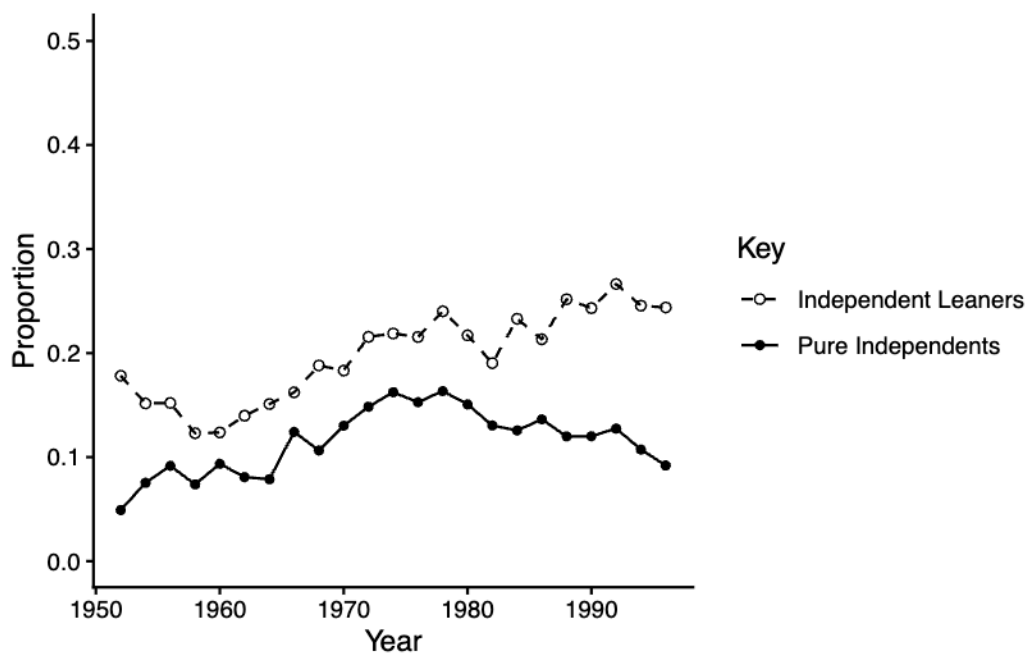
    name = "Key"
  ) +
  scale_linetype_manual(
    values = c("strong" = "solid", "pure" = "solid", "weak" = "dashed", "lean" = "dashed"),
    labels = c("lean" = "Independent Leaners", "pure" = "Pure Independents", "strong" = "Str
    name = "Key"
  ) +
  scale_y_continuous(limits = c(0, 0.5), breaks = seq(0, 0.5, by = 0.1)) +
  theme_classic() +
  geom_line() +
  geom_point()

```

```

#Check the graph
bottom_ggplot

```



```

#Build the graph (party ID Data) that will go on the top
top_ggplot <- ggplot(party_data, aes(x = VCF0004, y = prop, group = VCF0301_r, linetype = VCF0301_r)) +
  labs(x = "Year",
       y = "Proportion",
       title = "Figure 1: The Distribution of Party Identification, 1952-1996") +
  scale_shape_manual(
    values = c("strong" = 16, "pure" = 16, "weak" = 1, "lean" = 1),

```

```

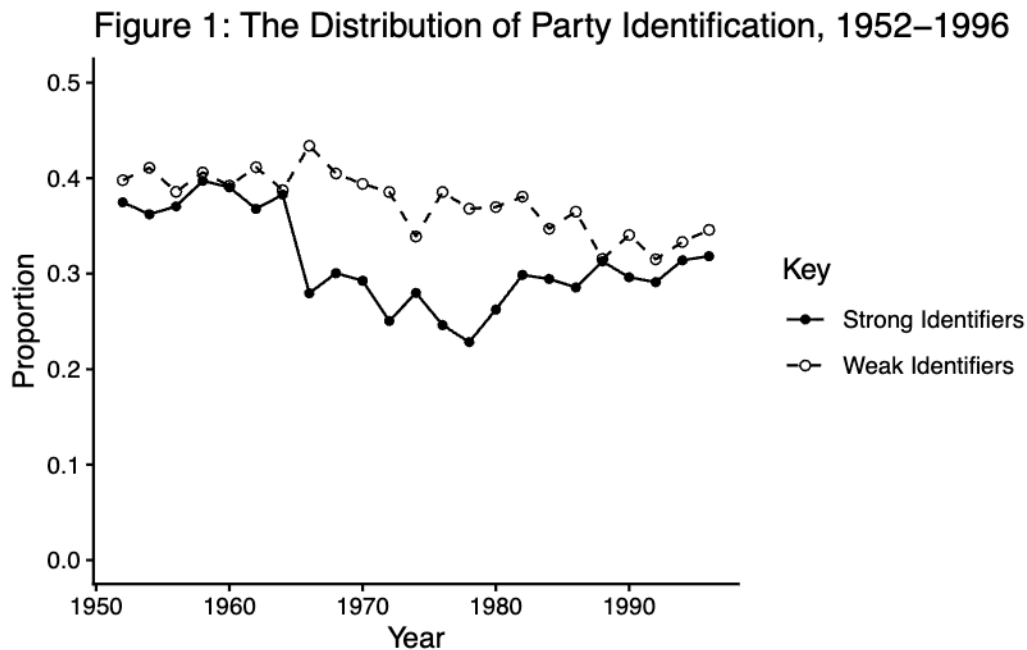
    labels = c("lean" = "Independent Leaners", "pure" = "Pure Independents", "strong" = "Str
    name = "Key"
  ) +
  scale_linetype_manual(
    values = c("strong" = "solid", "pure" = "solid", "weak" = "dashed", "lean" = "dashed"),
    labels = c("lean" = "Independent Leaners", "pure" = "Pure Independents", "strong" = "Str
    name = "Key"
  ) +
  scale_y_continuous(limits = c(0, 0.5), breaks = seq(0, 0.5, by = 0.1)) +
  theme_classic() +
  geom_line() +
  geom_point()

```

```

#Check that graph
top_ggplot

```



```

#Install Patchwork to Merge Graphs
library(patchwork)

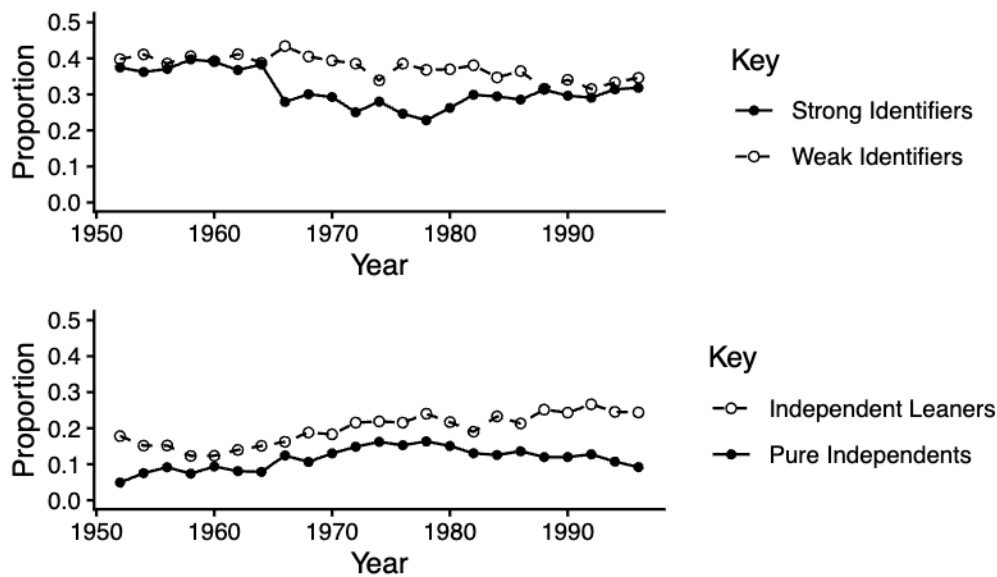
```

```

#Stack the Plots in Patchwork
top_ggplot/bottom_ggplot

```

Figure 1: The Distribution of Party Identification, 1952–1996



#Question 2

#Now I have to extend this analysis to recent years. To do this, I will run my old code, but

```
PartyIDyearly_r <- PartyIDyearly|>
```

```
mutate(VCF0301_r = case_when(
  VCF0301 %in% c(1, 7) ~ "strong",
  VCF0301 %in% c(2, 6) ~ "weak",
  VCF0301 %in% c(3, 5) ~ "lean",
  VCF0301 %in% c(4) ~ "pure"
))
```

#Filter Out the NA Data

```
NoNAPartyIDyearly_r <- PartyIDyearly_r |>
  filter(!is.na(VCF0301_r))
```

#Group by Year and Category, then Summarise and Find Proportions

```
yearlyprop2 <- NoNAPartyIDyearly_r |>
  group_by(VCF0004, VCF0301_r) |>
  summarise(countID = n()) |>
  mutate(prop = countID / sum(countID))
```

`summarise()` has regrouped the output.

```
i Summaries were computed grouped by VCF0004 and VCF0301_r.
i Output is grouped by VCF0004.
i Use `summarise(.groups = "drop_last")` to silence this message.
i Use `summarise(.by = c(VCF0004, VCF0301_r))` for per-operation grouping
  (`?dplyr::dplyr_by`) instead.
```

```
#Create New Columns Grouping by Party or Independent for the Two Graphs
yearlyprop2 <- yearlyprop2 |>
  mutate(dualgraph = case_when(
    VCF0301_r %in% c("strong", "weak") ~ "party",
    VCF0301_r %in% c("lean", "pure") ~ "independent"
  ))
```

```
#Begin Extended Stacked Plot
indep_data2 <- yearlyprop2 |> filter(dualgraph == "independent")
party_data2 <- yearlyprop2 |> filter(dualgraph == "party")
```

```
#Build 2nd Bottom Plot
bottom_ggplot2 <- ggplot(indep_data2, aes(x = VCF0004, y = prop, group = VCF0301_r, linetype =
  labs(x = "Year",
    y = "Proportion") +
  scale_shape_manual(
    values = c("strong" = 16, "pure" = 16, "weak" = 1, "lean" = 1),
    labels = c("lean" = "Independent Leaners", "pure" = "Pure Independents", "strong" = "Str
    name = "Key"
  ) +
  scale_linetype_manual(
    values = c("strong" = "solid", "pure" = "solid", "weak" = "dashed", "lean" = "dashed"),
    labels = c("lean" = "Independent Leaners", "pure" = "Pure Independents", "strong" = "Str
    name = "Key"
  ) +
  scale_y_continuous(limits = c(0, 0.5), breaks = seq(0, 0.5, by = 0.1)) +
  theme_classic() +
  geom_line() +
  geom_point()
```

```
#Build 2nd Top Plot and Change the Title
top_ggplot2 <- ggplot(party_data2, aes(x = VCF0004, y = prop, group = VCF0301_r, linetype =
  labs(x = "Year",
    y = "Proportion",
    title = "Figure 1: The Distribution of Party Identification, 1948-2020") +
  scale_shape_manual(
```



```

    values = c("strong" = 16, "pure" = 16, "weak" = 1, "lean" = 1),
    labels = c("lean" = "Independent Leaners", "pure" = "Pure Independents", "strong" = "Str
    name = "Key"
  ) +
  scale_linetype_manual(
    values = c("strong" = "solid", "pure" = "solid", "weak" = "dashed", "lean" = "dashed"),
    labels = c("lean" = "Independent Leaners", "pure" = "Pure Independents", "strong" = "Str
    name = "Key"
  ) +
  scale_y_continuous(limits = c(0, 0.5), breaks = seq(0, 0.5, by = 0.1)) +
  theme_classic() +
  geom_line() +
  geom_point()

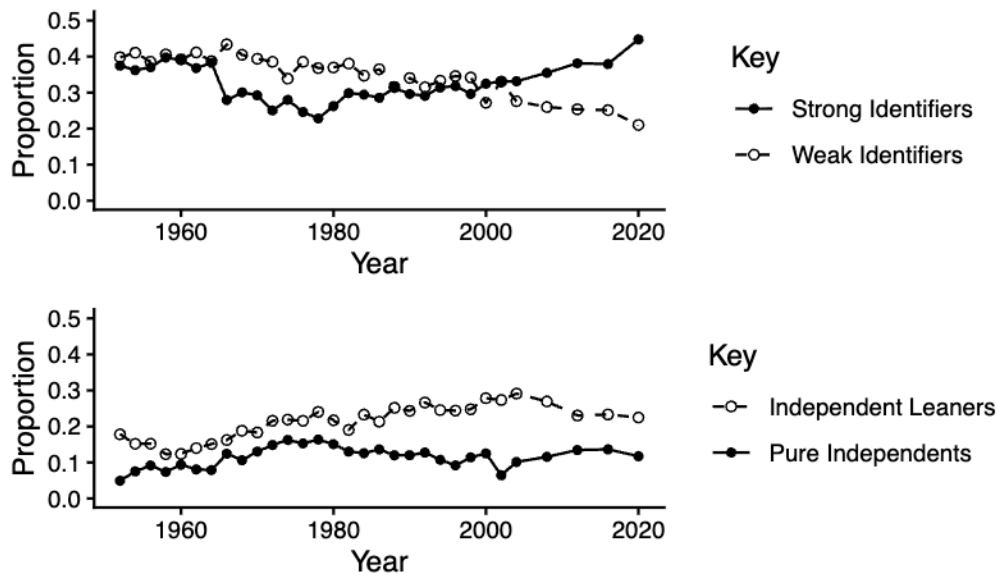
```

```

#Stack the Plots in Patchwork
top_ggplot2/bottom_ggplot2

```

Figure 1: The Distribution of Party Identification, 1948–2020



#Question 3

```

#Read in the Cumulative American National Election Studies (ANES) File from 2024
my2024data <- read_dta('/Users/brynsuiston/Documents/Data Wrangling Library/Week 7 Homework 1

```

```
#Create a Year Variable Because It's Missing
my2024data <- my2024data |>
  mutate(VCF0004 = 2024)
```

```
#Filter 2024 Data by Variables of Interest, V241226 and VCF0004
PartyID2024 <- c("V241227x", "VCF0004")
```

```
#Create Data Set for the 2024 Variable of Interest
PartyID2024 <- my2024data [, PartyID2024]
```

```
#Recode the PartyID Identifiers so We Can Get Either Strong or Weak, Pure or Lean, Checking
PartyID2024_r <- PartyID2024|>
  mutate(VCF0301_r = case_when(
    V241227x %in% c(1, 7) ~ "strong",
    V241227x %in% c(2, 6) ~ "weak",
    V241227x %in% c(3, 5) ~ "lean",
    V241227x %in% c(4) ~ "pure",
  ))
```

```
#Filter out the NA Variables
NoNAPartyID2024_r <- PartyID2024_r |>
  filter(!is.na(VCF0301_r))
data_2024 <- NoNAPartyID2024_r |>
  select(VCF0004, VCF0301_r)
```

```
#Double Check to Make Sure Everything is Sorted and the NAs are Filtered Out
table(data_2024$VCF0301_r, useNA = "ifany")
```

lean	pure	strong	weak
1430	380	2480	1193

```
#Make Sure Everything is Recoded From the Start
data_cdf <- PartyIDyearly |>
  filter(VCF0004 %in% 1952:2020) |>
  mutate(VCF0301_r = case_when(
    VCF0301 %in% c(1, 7) ~ "strong",
    VCF0301 %in% c(2, 6) ~ "weak",
    VCF0301 %in% c(3, 5) ~ "lean",
    VCF0301 %in% c(4) ~ "pure"
```

```

)) |>
filter(!is.na(VCF0301_r)) |>
select(VCF0004, VCF0301_r)

```

```

#Stack the Data Frames
combined <- bind_rows(data_cdf, data_2024)

```

```

#Group by Year and Category, then Summarise and Find Proportions
yearlypropw2024 <- combined |>
  group_by(VCF0004, VCF0301_r) |>
  summarise(countID = n()) |>
  mutate(prop = countID / sum(countID))

```

`summarise()` has regrouped the output.
 i Summaries were computed grouped by VCF0004 and VCF0301_r.
 i Output is grouped by VCF0004.
 i Use `summarise(.groups = "drop\_last")` to silence this message.
 i Use `summarise(.by = c(VCF0004, VCF0301\_r))` for per-operation grouping
 (`?dplyr::dplyr\_by`) instead.

```

#Create New Columns Grouping by Party or Independent for the Two Graphs Again
yearlypropw20242 <- yearlypropw2024 |>
  mutate(dualgraph = case_when(
    VCF0301_r %in% c("strong", "weak") ~ "party",
    VCF0301_r %in% c("lean", "pure") ~ "independent"
  ))

```

```

#Seperate the Two Groups
indep_data2024 <- yearlypropw20242 |> filter(dualgraph == "independent")
party_data2024 <- yearlypropw20242 |> filter(dualgraph == "party")

```

```

#Build the Two Graphs
bottom_ggplot2024 <- ggplot(indep_data2024, aes(x = VCF0004, y = prop, group = VCF0301_r, lin
  labs(x = "Year",
        y = "Proportion") +
  scale_shape_manual(
    values = c("strong" = 16, "pure" = 16, "weak" = 1, "lean" = 1),
    labels = c("lean" = "Independent Leaners", "pure" = "Pure Independents", "strong" = "Str
    name = "Key"
  ) +

```

```

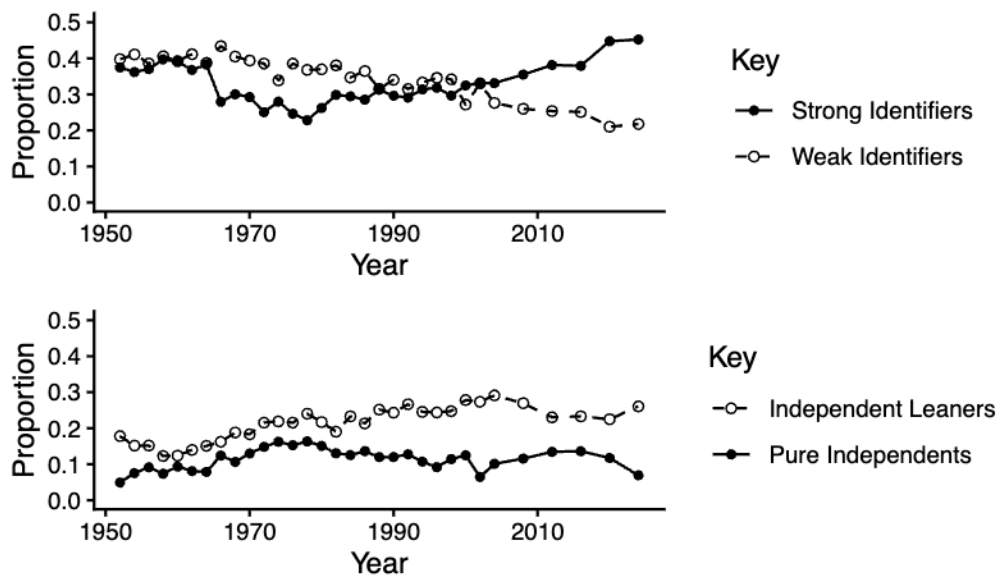
scale_linetype_manual(
  values = c("strong" = "solid", "pure" = "solid", "weak" = "dashed", "lean" = "dashed"),
  labels = c("lean" = "Independent Leaners", "pure" = "Pure Independents", "strong" = "Str
  name = "Key"
) +
scale_y_continuous(limits = c(0, 0.5), breaks = seq(0, 0.5, by = 0.1)) +
theme_classic() +
geom_line() +
geom_point()

top_ggplot2024 <- ggplot(party_data2024, aes(x = VCF0004, y = prop, group = VCF0301_r, linety
  labs(x = "Year",
       y = "Proportion",
       title = "Figure 1: The Distribution of Party Identification, 1952-2024") +
scale_shape_manual(
  values = c("strong" = 16, "pure" = 16, "weak" = 1, "lean" = 1),
  labels = c("lean" = "Independent Leaners", "pure" = "Pure Independents", "strong" = "Str
  name = "Key"
) +
scale_linetype_manual(
  values = c("strong" = "solid", "pure" = "solid", "weak" = "dashed", "lean" = "dashed"),
  labels = c("lean" = "Independent Leaners", "pure" = "Pure Independents", "strong" = "Str
  name = "Key"
) +
scale_y_continuous(limits = c(0, 0.5), breaks = seq(0, 0.5, by = 0.1)) +
theme_classic() +
geom_line() +
geom_point()

#Stack the Plots in Patchwork
top_ggplot2024/bottom_ggplot2024

```

Figure 1: The Distribution of Party Identification, 1952–2024



#Question 4

#I need to create a data set that separates out two different columns for election years and
mpcombined <- combined |>

```
mutate(election_type = case_when(
  VCF0004 %% 4 == 0 ~ "presidential",
  VCF0004 %% 4 == 2 ~ "midterm"
))
```

#Now I will use the grouping function and the proportion functions

```
mpcombinedwprop <- mpcombined |>
  filter(!is.na(election_type)) |>
  group_by(VCF0004, election_type, VCF0301_r) |>
  summarise(count = n()) |>
  mutate(prop = count / sum(count))
```

`summarise()` has regrouped the output.

i Summaries were computed grouped by VCF0004, election_type, and VCF0301_r.

i Output is grouped by VCF0004 and election_type.

i Use `summarise(.groups = "drop\_last")` to silence this message.

i Use `summarise(.by = c(VCF0004, election\_type, VCF0301\_r))` for per-operation grouping (`?dplyr::dplyr\_by`) instead.

```

#Create the Graphs
mpcombinedwprop_r <- mpcombinedwprop |>
  mutate(dualgraph = case_when(
    VCF0301_r %in% c("strong", "weak") ~ "party",
    VCF0301_r %in% c("lean", "pure") ~ "independent"
  ))

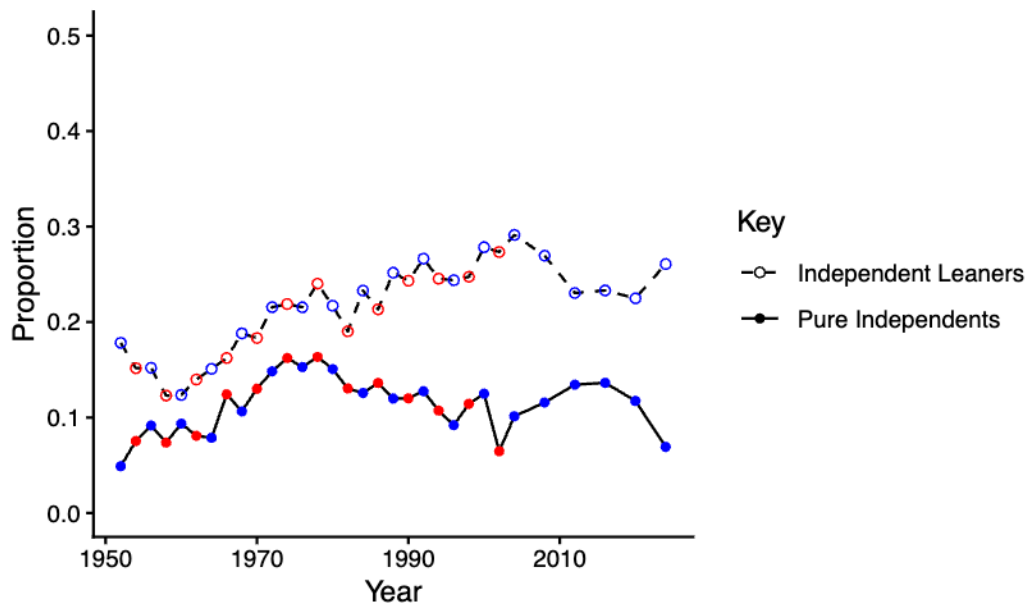
indep_datamp <- mpcombinedwprop_r |> filter(dualgraph == "independent")
party_datamp <- mpcombinedwprop_r |> filter(dualgraph == "party")

top_ggplotmp <- ggplot(indep_datamp, aes(x = VCF0004, y = prop, group = VCF0301_r, linetype =
  labs(x = "Year",
    y = "Proportion",
    title = "Figure 1: The Distribution of Party Identification, 1952-2024 by Election Type",
  scale_shape_manual(
    values = c("strong" = 16, "pure" = 16, "weak" = 1, "lean" = 1),
    labels = c("lean" = "Independent Leaners", "pure" = "Pure Independents", "strong" = "Str
    name = "Key"
  ) +
  scale_linetype_manual(
    values = c("strong" = "solid", "pure" = "solid", "weak" = "dashed", "lean" = "dashed"),
    labels = c("lean" = "Independent Leaners", "pure" = "Pure Independents", "strong" = "Str
    name = "Key"
  ) +
  scale_y_continuous(limits = c(0, 0.5), breaks = seq(0, 0.5, by = 0.1)) +
  guides(color = "none") +
  theme_classic() +
  geom_line() +
  geom_point(aes(color = election_type)) +
  scale_color_manual(
    values = c("presidential" = "blue", "midterm" = "red"),
    name = "Election type"
  )
)

```

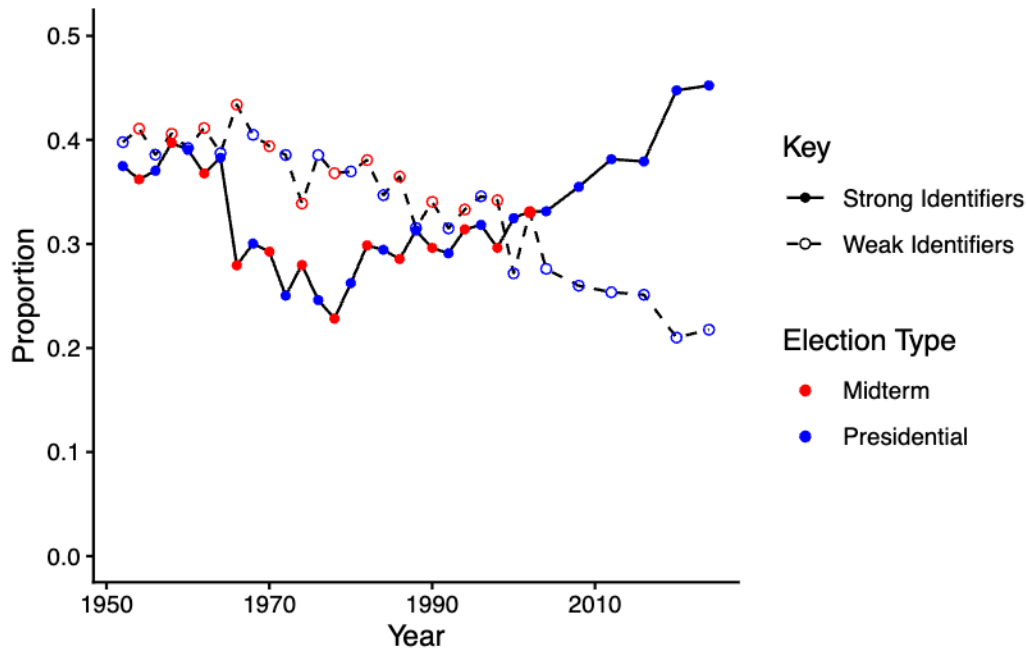
```
top_ggplotmp
```

Figure 1: The Distribution of Party Identification, 1952–2024 by



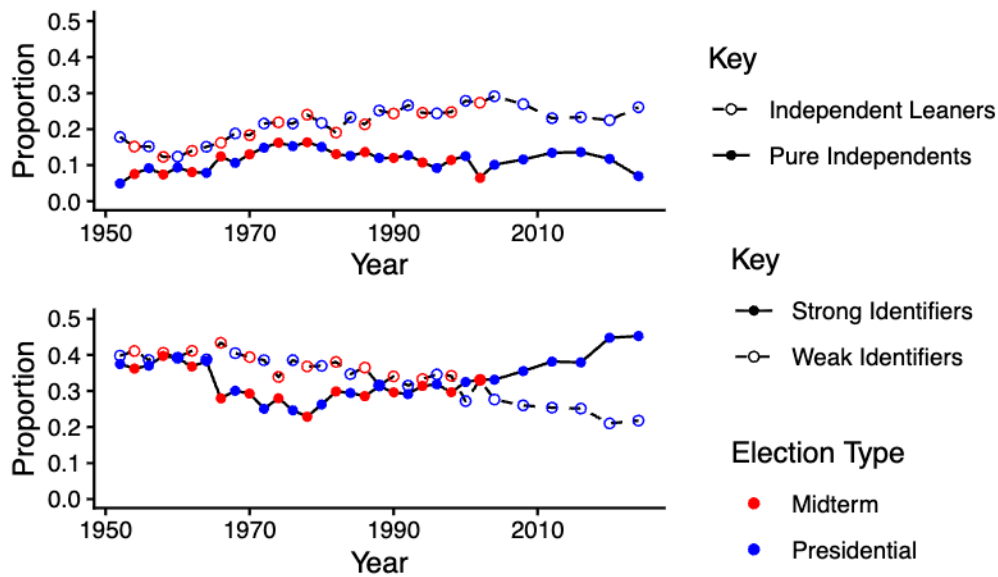
```
bottom_ggplotmp <- ggplot(party_datamp, aes(x = VCF0004, y = prop, group = VCF0301_r, linetype = election_type)) +
  labs(x = "Year",
       y = "Proportion") +
  scale_shape_manual(
    values = c("strong" = 16, "pure" = 16, "weak" = 1, "lean" = 1),
    labels = c("lean" = "Independent Leaners", "pure" = "Pure Independents", "strong" = "Strong", "weak" = "Weak"),
    name = "Key"
  ) +
  scale_linetype_manual(
    values = c("strong" = "solid", "pure" = "solid", "weak" = "dashed", "lean" = "dashed"),
    labels = c("lean" = "Independent Leaners", "pure" = "Pure Independents", "strong" = "Strong", "weak" = "Weak"),
    name = "Key"
  ) +
  scale_y_continuous(limits = c(0, 0.5), breaks = seq(0, 0.5, by = 0.1)) +
  theme_classic() +
  geom_line() +
  geom_point(aes(color = election_type)) +
  scale_color_manual(
    values = c("presidential" = "blue", "midterm" = "red"),
    labels = c("presidential" = "Presidential", "midterm" = "Midterm"),
    name = "Election Type"
  )
```

bottom_ggplotmp



top_ggplotmp/bottom_ggplotmp

Figure 1: The Distribution of Party Identification, 1952–2024 b



#Question 5

```
#I am now going to create my own graph from interesting variables from the data set
myvaribs <- c("VCF0748", "VCF0004", "VCF0824")
myvaribsyearly <- mydata[, myvaribs]
```

```
#Recode the Day Of/Early Variable
myvaribsyearly_r1 <- myvaribsyearly|>
  mutate(VCF0748_r = case_when(
    VCF0748 %in% c(1) ~ "On election day",
    VCF0748 %in% c(5) ~ "Some time before"
  ))
```

```
#Recode the ID Variable
myvaribsyearly_r2 <- myvaribsyearly_r1 |>
  mutate (VCF0824_r = case_when(
    VCF0824 %in% c(1) ~ "liberal",
    VCF0824 %in% c(2) ~ "lean liberal",
    VCF0824 %in% c(3) ~ "moderate",
    VCF0824 %in% c(4) ~ "lean conservative",
    VCF0824 %in% c(5) ~ "conservative"
  ))
```

```
#Filter Out the NA Data From Election
NoNAvaribsyearly_r <- myvaribsyearly_r2 |>
  filter(!is.na(VCF0748_r))
```

```
#Filter Out the NA Data From ID
NoNAvaribsyearly_r2 <- NoNAvaribsyearly_r |>
  filter(!is.na(VCF0824_r))
```

```
#Now I will use the grouping function and the proportion functions
myvaribsfinal1 <- NoNAvaribsyearly_r2 |>
  group_by(VCF0004, VCF0824_r, VCF0748_r) |>
  summarise(count = n()) |>
  mutate(prop = count / sum(count))
```

`summarise()` has regrouped the output.

i Summaries were computed grouped by VCF0004, VCF0824_r, and VCF0748_r.

i Output is grouped by VCF0004 and VCF0824_r.

i Use `summarise(groups = "drop\_last")` to silence this message.

i Use ``summarise(.by = c(VCF0004, VCF0824_r, VCF0748_r))`` for per-operation grouping (``?dplyr::dplyr_by``) instead.

```
ideology_totals <- myvaribsfinal1 |>
  group_by(VCF0824_r) |>
  summarise(total = sum(count))

bars_data <- myvaribsfinal1 |>
  group_by(VCF0824_r, VCF0748_r) |>
  summarise(count = sum(count))
```

``summarise()`` has regrouped the output.

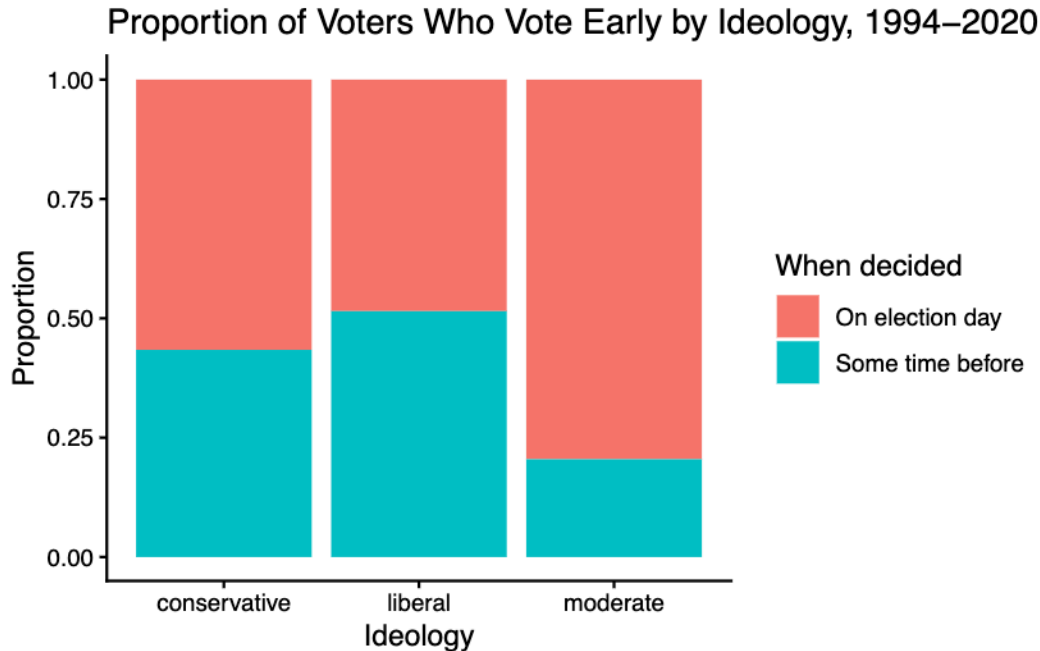
i Summaries were computed grouped by VCF0824_r and VCF0748_r.

i Output is grouped by VCF0824_r.

i Use ``summarise(.groups = "drop_last")`` to silence this message.

i Use ``summarise(.by = c(VCF0824_r, VCF0748_r))`` for per-operation grouping (``?dplyr::dplyr_by``) instead.

```
#Create Graph
ggplot(bars_data, aes(x = VCF0824_r, y = count, fill = VCF0748_r)) +
  geom_col(position = "fill") +
  labs(x = "Ideology",
       y = "Proportion",
       fill = "When decided",
       title = "Proportion of Voters Who Vote Early by Ideology, 1994-2020") +
  theme_classic()
```



#Response Question Answers 1. My first problem that I ran into was learning how to stack the plots. As you can see in my code, I thought I could make one graph initially, but I learned that I had to build the plots individually and then stack them with patchwork. Another tricky aspect was merging the 2024 data set. I had to do a fair amount of research to figure out how to combine them, making sure the rows and the columns were going to line up smoothly.

2. In Figure 1, Bartels found that the number of independents, whether “pure” or “leaners” has decreased, and in the same vein, party identification has gone up. Most people remain either “weakly” identified with a party or “leaners. In Figure 2, Party identification as a whole remains strongest amongst voters as compared to non-voters.
3. Since 1996, you can see that strong identifiers have significantly swung upwards, while pure independents have dipped. To me, this demonstrates that there has been increased polarization in the United States politically. I think this actually supports Bartel’s findings that the “decline of parties” simply isn’t happening, but identification is in fact trending in the opposite direction.
4. The inclusion of the midterm was interesting, but we run into the road block of the missing data. Overall, it looks like it consistently swings opposite of the presidential election. When one increases, the other decreases and vice versa. However, I am doubtful it changes anything about the overall analysis.